HORIZON MEDICAL — Software Documentation (IEC 62304)

Documento: HM-SWD-001
Versión: 1.0
Fecha: 2026-05-16
Referencia: IEC 62304:2006+AMD1:2015, FDA Software Guidance

Tabla de Contenidos

1. Software Safety Classification

1.1 Classification per IEC 62304

Software System / Item	Safety Class	Justification
System: Horizon Medical Holter ECG Software	**Class C**	Software whose failure can result in serious injury or death (misdetection of VT/VF)
Firmware (nRF52832)	Class C	Acquires and processes ECG signal; failure could result in loss or corruption of diagnostic data
Backend / Cloud Processing	Class C	Processes and stores clinical data; hosts AI engine for arrhythmia detection
AI Module (CNN-LSTM)	Class C	Detects potentially lethal arrhythmias; false negative could contribute to serious harm
Mobile App (iOS/Android)	Class B	Displays ECG and alerts; not the primary diagnostic interface (dashboard is); failure results in temporary loss of patient-facing alerts
Dashboard Web (Clinical)	Class B	Primary interface for clinician review; failure delays but does not prevent diagnosis (data is stored and can be accessed later)

1.2 FDA Level of Concern

Software Item	Level of Concern	Justification
Firmware	Major	Directly acquires physiological signals used for clinical decisions
Backend + AI	Major	AI detection of life-threatening arrhythmias
Mobile App	Moderate	Patient-facing visualization; not primary diagnostic tool
Dashboard	Moderate	Clinician interface; data available through other means if dashboard fails

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2. Software Development Plan

2.1 Development Lifecycle Model

Model: V-Model (adapted for agile sprints within phases)

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User Needs ──────────────────────────────── Validation
     │                                          ▲
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Software Requirements ──────────────────── System Testing
     │                                          ▲
     ▼                                          │
Architecture Design ──────────────────── Integration Testing
     │                                          ▲
     ▼                                          │
Detailed Design ──────────────────────── Unit Testing
     │                                   ▲
     ▼                                   │
     └──────── Coding ──────────────────┘
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2.2 Development Activities per IEC 62304

Activity	Class A	Class B	Class C	Horizon Applies
Software development planning	✅	✅	✅	✅
Software requirements analysis	✅	✅	✅	✅
Software architectural design	—	✅	✅	✅
Software detailed design	—	—	✅	✅
Software unit implementation	✅	✅	✅	✅
Software unit verification	—	—	✅	✅
Software integration and integration testing	—	✅	✅	✅
Software system testing	✅	✅	✅	✅
Software release	✅	✅	✅	✅

2.3 Development Tools

Tool	Purpose	Version	Validation Status
Git	Version control	Latest	Tool qualification not required (well-established)
Jira	Requirements management, traceability	Cloud	Validated for intended use
Segger Embedded Studio / VS Code	Firmware IDE	[Ver]	Tool qualification per IEC 62304
GCC ARM	Firmware compiler	[Ver]	Compiler validation testing performed
PyTorch / TensorFlow	AI model development	[Ver]	Validated for model training
React Native CLI	Mobile app build	[Ver]	Build output verified
Jest / Pytest / Unity (C)	Unit testing frameworks	[Ver]	N/A (test tools)
SonarQube	Static code analysis	[Ver]	N/A (analysis tool)
Jenkins / GitHub Actions	CI/CD	[Ver]	Pipeline validation performed

2.4 Configuration Management

Element	Method
Source code	Git repository with branch protection, code review (PR) required
Requirements	Jira with traceability links
Documents	Document control system (QMS)
Build artifacts	CI/CD with artifact versioning
Release management	Git tags, release notes, checksum verification

3. Software Requirements Specification

3.1 Firmware Requirements (nRF52832)

Req ID	Requirement	Type	Priority	Verification
FW-REQ-001	The firmware shall acquire ECG data from ADS1298 at 500 Hz per channel, 24-bit resolution	Functional	Mandatory	Unit test + bench test
FW-REQ-002	The firmware shall apply bandpass filtering (0.05–150 Hz) to the raw ECG signal	Functional	Mandatory	Unit test + frequency response test
FW-REQ-003	The firmware shall apply 50/60 Hz notch filter	Functional	Mandatory	Unit test
FW-REQ-004	The firmware shall detect electrode disconnection via impedance measurement	Functional	Mandatory	Unit test + bench test
FW-REQ-005	The firmware shall transmit ECG data via BLE to the paired mobile device	Functional	Mandatory	Integration test
FW-REQ-006	The firmware shall buffer ECG data locally when BLE is disconnected (≥8 hours)	Functional	Mandatory	Unit test
FW-REQ-007	The firmware shall monitor battery level and generate alerts at ≤20% and ≤5%	Functional	Mandatory	Unit test

Req ID	Requirement	Type	Priority	Verification
FW-REQ-008	The firmware shall support secure OTA updates with dual-bank and rollback	Functional	Mandatory	Integration test
FW-REQ-009	The firmware shall use BLE Secure Connections with AES-128 encryption	Security	Mandatory	Security test
FW-REQ-010	The firmware shall implement watchdog timer for crash recovery	Reliability	Mandatory	Unit test

3.2 Backend Requirements

Req ID	Requirement	Type	Priority
BE-REQ-001	The backend shall receive ECG data from mobile apps via HTTPS/TLS 1.3	Functional	Mandatory
BE-REQ-002	The backend shall store ECG data with AES-256 encryption at rest	Security	Mandatory
BE-REQ-003	The backend shall process ECG data through the AI module within 30 seconds of receipt	Performance	Mandatory
BE-REQ-004	The backend shall implement OAuth 2.0 authentication for all API endpoints	Security	Mandatory
BE-REQ-005	The backend shall support RBAC (clinician, admin, patient roles)	Security	Mandatory
BE-REQ-006	The backend shall maintain audit logs for all data access events	Security	Mandatory
BE-REQ-007	The backend shall achieve ≥99.9% uptime (SLA)	Availability	Mandatory
BE-REQ-008	The backend shall implement rate limiting to prevent API abuse	Security	Mandatory

3.3 AI Module Requirements

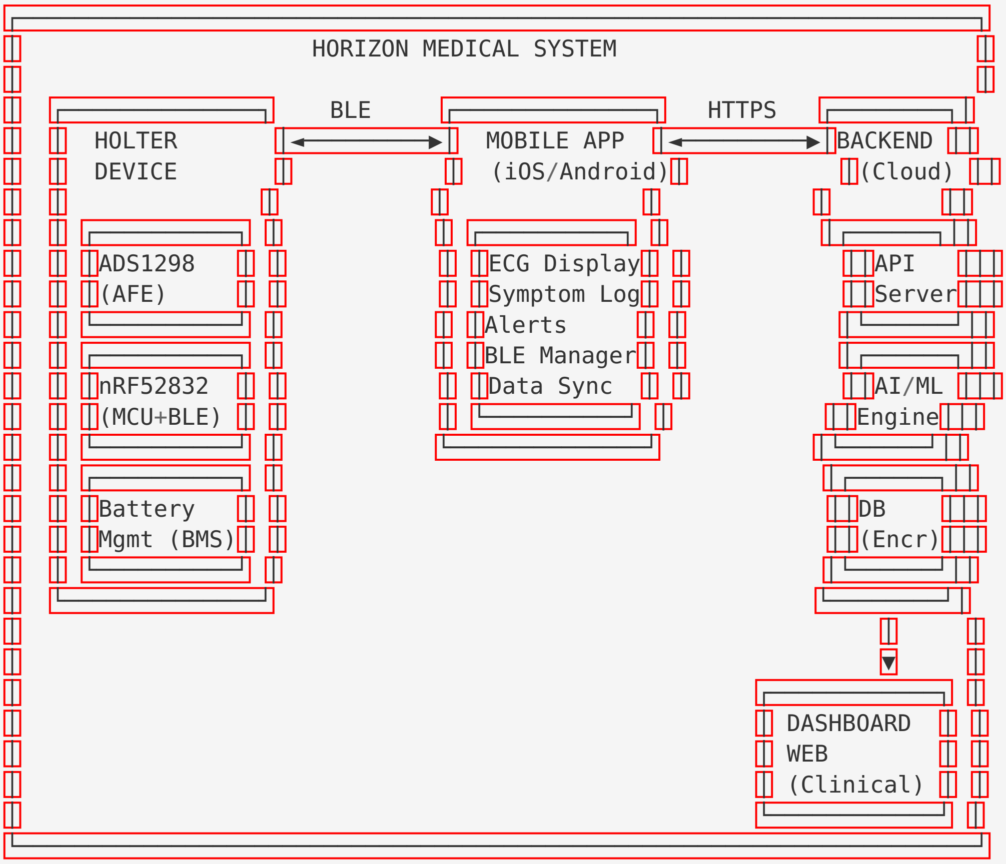
Req ID	Requirement	Type	Priority
AI-REQ-001	The AI module shall detect AF with sensitivity ≥95% and specificity ≥95%	Performance	Mandatory
AI-REQ-002	The AI module shall detect VT with sensitivity ≥95% and specificity ≥98%	Performance	Mandatory
AI-REQ-003	The AI module shall classify ECG segments into defined arrhythmia categories	Functional	Mandatory
AI-REQ-004	The AI module shall provide confidence scores for each classification	Functional	Mandatory
AI-REQ-005	The AI module shall process a 10-second ECG segment in <5 seconds	Performance	Mandatory
AI-REQ-006	The AI module shall be a locked algorithm (no continuous learning in production)	Regulatory	Mandatory
AI-REQ-007	The AI module shall log all inference inputs and outputs for auditability	Regulatory	Mandatory

3.4 Mobile App Requirements

Req ID	Requirement	Type	Priority
APP-REQ-001	The app shall display real-time ECG wave-form from the Holter device	Functional	Mandatory
APP-REQ-002	The app shall display battery level of the Holter device	Functional	Mandatory
APP-REQ-003	The app shall allow the patient to log symptoms with timestamp	Functional	Mandatory
APP-REQ-004	The app shall display alerts received from the backend	Functional	Mandatory
APP-REQ-005	The app shall trans-mit ECG data to the backend when Inter-net is available	Functional	Mandatory
APP-REQ-006	The app shall store ECG data locally when offline and sync when reconnected	Functional	Mandatory
APP-REQ-007	The app shall support iOS 15+ and Android 12+	Compatibility	Mandatory

4. Software Architecture

4.1 System Architecture



4.2 Data Flow

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Electrodes → ADS1298 (ADC 24-bit, 500Hz) → nRF52832 (filtering, buffering)
    → BLE → Mobile App (display, cache) → HTTPS/TLS 1.3 → Backend API
    → AI Engine (CNN-LSTM inference) → Database (encrypted storage)
    → Dashboard (clinical review) → PDF Report
```

4.3 Software Decomposition

Software System	Software Items	SOUP Items
Firmware	ECG acquisition, digital filter, BLE stack, OTA updater, battery manager, electrode monitor	Zephyr RTOS, Nordic SDK, mbed TLS
Backend	API server, data pipeline, auth service, notification service	Node.js/Python runtime, Express/FastAPI, PostgreSQL, Redis
AI Module	Preprocessing, CNN-LSTM model, inference engine, confidence scorer	PyTorch/TensorFlow, NumPy, SciPy
Mobile App	BLE manager, ECG renderer, data sync, UI components	React Native, BLE library
Dashboard	Auth module, ECG viewer, report generator, patient manager	React.js, Chart.js/D3.js

5. Software Detailed Design

5.1 Firmware Modules

Module	Description	Interfaces	Safety Relevance
`ecg_acquisition`	Configures ADS1298, reads samples via SPI, manages sampling rate	SPI to ADS1298, internal queue	Class C — primary data source
`digital_filter`	Implements IIR bandpass (0.05–150 Hz) and FIR notch (50/60 Hz)	Internal data pipeline	Class C — signal integrity
`ble_comm`	Manages BLE advertising, connection, data transmission	BLE stack API	Class C — data transmission
`data_buffer`	Circular buffer for offline storage (flash memory)	Internal storage API	Class C — data preservation
`electrode_monitor`	Monitors electrode impedance, detects disconnection	ADC channel, alert system	Class C — data quality
`battery_manager`	Monitors voltage, manages power states, generates alerts	ADC, GPIO, alert system	Class B — operational continuity
`ota_updater`	Dual-bank firmware update with rollback	BLE, flash memory	Class C — device integrity
`watchdog`	Hardware watchdog timer, system health monitoring	WDT peripheral	Class B — reliability

5.2 AI Module Components

Component	Description	Input	Output
`preprocessor`	Resamples, normalizes, segments ECG into 10-second windows	Raw ECG data (3-channel, 500 Hz)	Normalized segments (10s × 3ch × 5000 samples)
`feature_extractor` (CNN)	Convolutional layers extract spatial features from ECG segments	Preprocessed segments	Feature vectors
`sequence_analyzer` (LSTM)	LSTM layers analyze temporal patterns across segments	Feature vectors (sequential)	Temporal feature representation
`classifier`	Dense layers + softmax for arrhythmia classification	Temporal features	Class probabilities + confidence score
`post_processor`	Applies thresholds, aggregates results, generates alerts	Class probabilities	Arrhythmia labels, alert priority

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6. Software Unit Testing

6.1 Unit Testing Strategy

Software Item	Test Framework	Coverage Target	Approach
Firmware	Unity (C), CMock	≥80% line coverage	White-box testing of critical modules
Backend	Jest (Node.js) / Pytest (Python)	≥80% line coverage	White-box + boundary testing
AI Module	Pytest	≥70% line coverage + model tests	Unit tests + performance tests per class
Mobile App	Jest + React Native Testing Library	≥70% line coverage	Component + integration tests
Dashboard	Jest + React Testing Library	≥70% line coverage	Component tests

6.2 Critical Unit Tests (Firmware)

Test ID	Module	Test Description	Expected Result	Status
UT-FW-001	`digital_filter`	Verify bandpass filter with known sinusoidal inputs at 0.01, 0.05, 1, 50, 150, 200 Hz	Correct attenuation per specification	⬜
UT-FW-002	`digital_filter`	Verify notch filter attenuates 50 Hz by ≥40 dB	≥40 dB attenuation at 50 Hz	⬜
UT-FW-003	`electrode_monitor`	Verify disconnection detection when impedance > 50 kΩ	Alert generated within 5 seconds	⬜
UT-FW-004	`data_buffer`	Fill buffer to capacity, verify oldest data is overwritten (circular)	FIFO behavior confirmed	⬜
UT-FW-005	`battery_manager`	Simulate voltage at 20% and 5% thresholds	Correct alerts generated	⬜

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7. Software Integration Testing

7.1 Integration Test Cases

Test ID	Components	Test Description	Expected Result	Status
IT-001	Firmware → BLE → App	ECG data transmitted from device to app in real-time	ECG displayed on app with <2s latency	⬜
IT-002	App → Backend	ECG data uploaded from app to backend via HTTPS	Data received, validated, stored in DB	⬜
IT-003	Backend → AI Module	ECG data processed by AI engine	Arrhythmia classification returned within 30s	⬜
IT-004	Backend → Dashboard	Clinical data displayed on dashboard	Correct patient data, ECG, AI results shown	⬜
IT-005	Backend → App (alerts)	Alert notification sent from backend to app	Push notification received on app	⬜
IT-006	Firmware (offline) → App (reconnect) → Backend	Buffered data synced after BLE reconnection	All buffered data successfully uploaded	⬜
IT-007	OTA Update flow	Firmware update via BLE from app	Firmware updated successfully, rollback tested	⬜

8. Software System Testing

8.1 System Test Cases

Test ID	Description	Acceptance Criteria	Status
ST-001	End-to-end: 48-hour continuous monitoring simulation	No data loss, all arrhythmias detected, battery sustains	⬜
ST-002	Performance under maximum load (concurrent users on dashboard)	Response time <3s with 50 concurrent users	⬜
ST-003	Offline mode: device operates without Internet for 8 hours, then syncs	All data recovered and processed correctly	⬜
ST-004	Security: penetration testing of API, BLE, dashboard	No critical/high vulnerabilities	⬜
ST-005	Stress test: rapid BLE connect/disconnect cycles	No crash, no data corruption	⬜
ST-006	Compatibility: test on 5 Android + 3 iOS devices	App functions correctly on all devices	⬜
ST-007	Recovery: backend crash and restart during data processing	No data loss, processing resumes	⬜
ST-008	AI performance regression test with golden dataset	Metrics within specification (Se ≥95% AF, Se ≥95% VT)	⬜

9. Software Validation

9.1 Validation Activities

Activity	Method	Acceptance Criteria
Clinical validation	Comparative study vs. standard Holter	Diagnostic concordance ≥90%
AI clinical validation	Independent dataset evaluation	Se/Sp within specification
Usability validation	Summative study (patients + clinicians)	No critical use errors
Home use validation	Real-world deployment in patient homes	Task completion ≥95%

9.2 Traceability Matrix Summary

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User Needs → Product Requirements → Software Requirements → Design → Code → Unit Tests
→ Integration Tests → System Tests → Validation
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Complete bidirectional traceability is maintained in [Jira/Requirements Management Tool].

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10. Software Maintenance

10.1 Maintenance Plan

Activity	Frequency	Responsible
Bug fix releases	As needed (severity-based SLA)	Development team
Security patches	As needed (CVSS-based priority)	Security team
Planned updates (features)	Quarterly	Product team
AI model updates	Annual (or as new data available)	AI team
SOUP/dependency updates	Quarterly review	Development team
Vulnerability monitoring	Continuous (SBOM-based)	Security team

10.2 Change Control for Software

All software changes follow the Design Change Control process:

1. Change Request submitted with impact assessment
2. Risk analysis of the change (ISO 14971)
3. Regression analysis to determine required re-testing
4. Implementation with code review
5. Regression testing execution
6. Release approval by Quality
7. Version increment and release notes

10.3 Problem Resolution

Severity	Response Time	Resolution Target
Critical (safety-related)	4 hours	24 hours (hotfix)
High (major functionality)	24 hours	1 week
Medium (minor functionality)	1 week	Next planned release
Low (cosmetic, minor)	2 weeks	Future release

Appendix: SOUP (Software of Unknown Provenance) List

SOUP Item	Version	Manufacturer	License	Risk Class	Anomaly List Reviewed
Zephyr RTOS	[Ver]	Zephyr Project	Apache 2.0	Class C	✅
Nordic nRF5 SDK	[Ver]	Nordic Semiconductor	BSD	Class C	✅
mbed TLS	[Ver]	ARM	Apache 2.0	Class C	✅
PyTorch	[Ver]	Meta/PyTorch	BSD	Class C	✅
React Native	[Ver]	Meta	MIT	Class B	✅
React.js	[Ver]	Meta	MIT	Class B	✅
Node.js	[Ver]	OpenJS Foundation	MIT	Class C	✅
PostgreSQL	[Ver]	PostgreSQL Global	PostgreSQL	Class C	✅

Control de Versiones:

Versión	Fecha	Autor	Descripción
1.0	2026-05-16	Ingeniería de Software	Creación inicial
